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ORIGINAL RESEARCH

Modifiable methodological and reporting practices are associated with reproducibility of health sciences research: a systematic review and evidence and gap map

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Abstract

Objectives: Map the evidence on factors (eg, research practices) associated with reproducibility of methods and results reported in health sciences research.

Study Design and Setting: Five bibliographic databases were searched from January 2000 to May 2023 with supplemental searches of high-impact journals and relevant records. We included health science records of observational, interventional, or knowledge synthesis studies reporting data on factors related to research reproducibility. Data were coded using inductive qualitative content analysis, and empirical evidence was synthesized with evidence and gap maps. Factors were categorized as modifiable or nonmodifiable; reproducibility outcomes were categorized as related to methods or results. Statistical tests of association between factors and reproducibility outcomes were summarized.

Results: Our review included 148 studies, primarily from biomedical/preclinical (n = 62) and clinical (n = 71) domains. Factors were classified into 12 modifiable (eg, sample size and power) and three nonmodifiable (eg, publication year) categories. Of 234 reported evaluations of factors, 76 (32%) assessed methodological reproducibility and 158 (68%) assessed results reproducibility. The most frequently reported factor was transparency and reporting (38 of 234 assessments). A total of 155 factors (66%) were evaluated for statistical associations with reproducibility outcomes. Statistical associations were most frequently conducted for analytical methods (24 of 26 reporting significance), sample size and power (21 of 23 reporting significance), and participants characteristics and study materials (10 of 12 reporting significance).

Conclusion: Several modifiable factors were associated with reproducibility of health sciences research and represent opportunities for intervention. Applying more stringent statistical testing procedures and thresholds, conducting appropriate sample size and power calculations, and improving transparency and completeness of reporting should be top priorities for improving reproducibility. Experimental studies to test interventions to improve reproducibility are needed.

Keywords: Reproducibility; Replicability; Meta-research; Research design and methods; Reporting practices; Research reporting guidelines; Evidence and gap map; Health sciences; Systematic review; Open science

1. Introduction

Scientific discovery has translated into dramatic improvements in human health. This progress is a result of evolutions in scientific inquiry and ways of thinking, the development of new technologies, and the progressive establishment of a robust knowledge base that informs health-care practices. While the progress of research is not linear, it must be reliable and valid to advance science and improve human health. Nevertheless, there are growing

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Plain Language Summary

Many research findings in medicine and health cannot be reproduced by other researchers. This makes it harder to know what evidence to trust when making patient care and health policy decisions. We systematically reviewed 148 studies that examined how specific research practices are linked to whether research findings can be reproduced. We found that several modifiable features of study design, analysis, and reporting were often associated with better reproducibility. Using larger sample sizes, applying more stringent statistical methods, and providing transparent, complete descriptions of methods and results were frequently linked to more reproducible findings. These results suggest that improving how studies are planned, analyzed, and reported may increase the trustworthiness and usefulness of health research.

concerns about the reliability and validity of many health sciences research studies.

Evaluating the reproducibility of research is one approach used in meta-science, the scientific investigation of science itself, to quantify the reliability and validity of research [1,2]. It has been estimated that a large proportion of findings in the health sciences cannot be reproduced [3−7]. Estimates of the reproducibility of research in the health sciences range from 35% to 75% [1,2,4,5,8,9]. The impacts of research that cannot be reproduced are substantial, including wasted research resources, increased research and health-care costs, and even loss of human life [5,7,9,10].

To address this “reproducibility crisis” in the health sciences, it is critical to identify factors that affect research reproducibility. Systematic reviews have examined factors associated with reproducibility in select health sciences fields [1,2,5,7,9,10]. However, there is a need to synthesize the available evidence for factors that impact the reproducibility of health sciences research more broadly to identify targets for intervention. We conducted a systematic review and evidence and gap map of health sciences research literature to identify and categorize modifiable and nonmodifiable factors associated with health sciences research reproducibility.

2. Methods

In this meta-research study, we reviewed studies about studies, whereby we examined empirical investigations of factors that may influence whether health sciences research is reproducible, rather than conducting new reproduction attempts of individual primary studies. We focused on two reproducibility outcomes—methodological reproducibility (the ability to repeat study procedures and data analyses) and results reproducibility (the ability to obtain corroborating results using the same or similar methods)—and we grouped the factors into modifiable and nonmodifiable categories.

2.1. Review design

The protocol for this systematic review and evidence and gap map was developed in accordance with the Cochrane Handbook for Guidelines on Conducting Systematic Reviews [11,12] and the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocols (PRISMA-P) checklist [13] and prospectively registered with the Open Science Framework (DOI:10.17605/OSF.IO/3FHD9) and PROSPERO (CRD42023399717). We followed the PRISMA reporting guidelines to present this work (Supplementary Table 1) [14].

2.2. Search strategy

The search strategy was developed in consultation with a health sciences librarian (DL) and validated by a second health sciences librarian (ZP) using the Peer Review of Electronic Search Strategies (PRESS) [15]. We searched five multidisciplinary databases with broad coverage of health sciences (MEDLINE, Embase, Web of Science, Cochrane Controlled Trials Registry, and Cochrane Database of Systematic Reviews) from January 01, 2000, to May 24, 2023, for articles published in English (Supplementary Table 2). We limited searches to starting in January 2000 to focus on contemporary reproducibility and meta-research (eg, CONSORT/PRISMA adoption). We restricted inclusion to English-language publications due to resource constraints for full-text translation; where available, we screened English abstracts of non-English records to gauge relevance.

Supplemental search strategies included: 1) keyword (“reproducibility” and “health science”) searching of select high-impact health sciences journals (New England Journal of Medicine, Lancet, Journal of the American Medical Association, Nature, Science, Cell); and 2) hand searching for articles on websites of scientific institutions that have previously published reports on research reproducibility or have ongoing reproducibility initiatives (e.g., National Institutes of Health, Academy of Medical Sciences, National Science Foundation). We screened reference lists of key meta-research overviews and included studies to corroborate database retrievals. The comprehensive search strategy is provided in Appendix 1.

2.3. Eligibility criteria

Papers were eligible for inclusion if they were health sciences research articles, general science articles that

What is new?

Key findings

- We synthesized evidence from 148 studies and identified 12 modifiable and 3 nonmodifiable factor categories associated with methodological and/or results reproducibility in health sciences research.
- Across the available literature, more stringent analytical/statistical practices and adequate sample size/power were most consistently associated with greater results reproducibility, while transparent and complete reporting was associated with greater methodological reproducibility.

What this adds to and what is known?

- Prior work has highlighted a reproducibility problem and field-specific evidence; this review provides a cross-health-sciences map of reproducibility-related factors and organizes them into actionable (modifiable) vs. structural (nonmodifiable) targets for intervention.

What is the implication and what should change now?

- Evidence remains concentrated in biomedical/preclinical and clinical research, with comparatively sparse empirical evaluation in health/social care services and population/public health domains.
- The current evidence base is predominantly observational; the next step is to run experimental studies of reproducibility-improving interventions and to expand meta-research in understudied domains that directly inform health system and population decisions.

explicitly indicated relevance to all scientific disciplines, or articles from specific fields of psychology that align with health sciences (clinical, developmental, health, and cognitive psychology), as well as behavioral neuroscience, psychopathology, psychophysiology, and psychotherapy [16,17]. Our objective was to map cross-cutting factor categories across the health sciences rather than within individual disciplines.

We included any primary research study reporting an observational or interventional study design, as well as knowledge syntheses. Studies were included if they reported data on at least one factor investigated in relation to methodological or results reproducibility. *Factors* were operationalized as modifiable or nonmodifiable aspects of study conduct, relating to individual-, study-, or institutional-level practices, methods, and processes that could impact the reproducibility of research methods or results. Reproducibility was operationalized by two mutually exclusive categories: 1) *methodological reproducibility*, defined as the ability to exactly repeat the methods, including study procedures and data analysis and 2) *results reproducibility*, defined as obtaining corroborating results using the same or similar methods (Supplementary Table 3) [18−21]. We also included studies that used surrogate measures for reproducibility (eg, Type I and Type II error rates) if they provided 1) an explicit statement that their aim was to investigate the reproducibility of research and 2) a rationale for their choice of surrogate measure.

For generalizability of findings, single studies that collected new data to reproduce the results of another single study were excluded. However, knowledge syntheses of multiple single reproduction attempts were included. Studies that reported successive measurements or experiments conducted by the same investigators in close temporal proximity using identical equipment and/or specimens were excluded since these studies were examining reliability or internal validity in the context of a single model. We excluded preprint records and conference abstracts. Categorization of study designs used in the review is provided in Supplementary Table 4.

2.4. Study screening and data abstraction

A random sample of 20 records was used to calibrate reviewer screening decisions. Repeat samples were pilot tested until 100% agreement was achieved. Thereafter, all titles and abstracts were reviewed independently and in duplicate by two reviewers (JK, SJM, or NB). Full text review followed by data abstraction was performed independently and in duplicate by two reviewers (JK, SJM), and disagreements were resolved by discussion and/or the involvement of a third reviewer (NB). We extracted one estimate for each factor category. Estimates were prioritized for extraction using the following decision rule: 1) study level summary estimates of reproducibility; 2) estimates based on the fewest conditional statements (eg, an estimate of the association of a factor and reproducibility derived from all studies examined as opposed to only studies reporting statistically significant associations); 3) estimates with the largest sample size; 4) adjusted estimates; and 5) unadjusted estimates.

Risk of bias was assessed using an adapted version of the QUalIty of Prognostic factor Studies (QUIPS) risk of bias tool and following the PROGnosis RESearch Strategy (PROGRESS) framework for prognostic research [22,23]. A subset of the team (JK, SJM, NB) calibrated the adapted QUIPS tool on 20 random records to achieve 100% agreement prior to two reviewers (JK, SJM), assessing the risk of bias independently and in duplicate for all included studies.

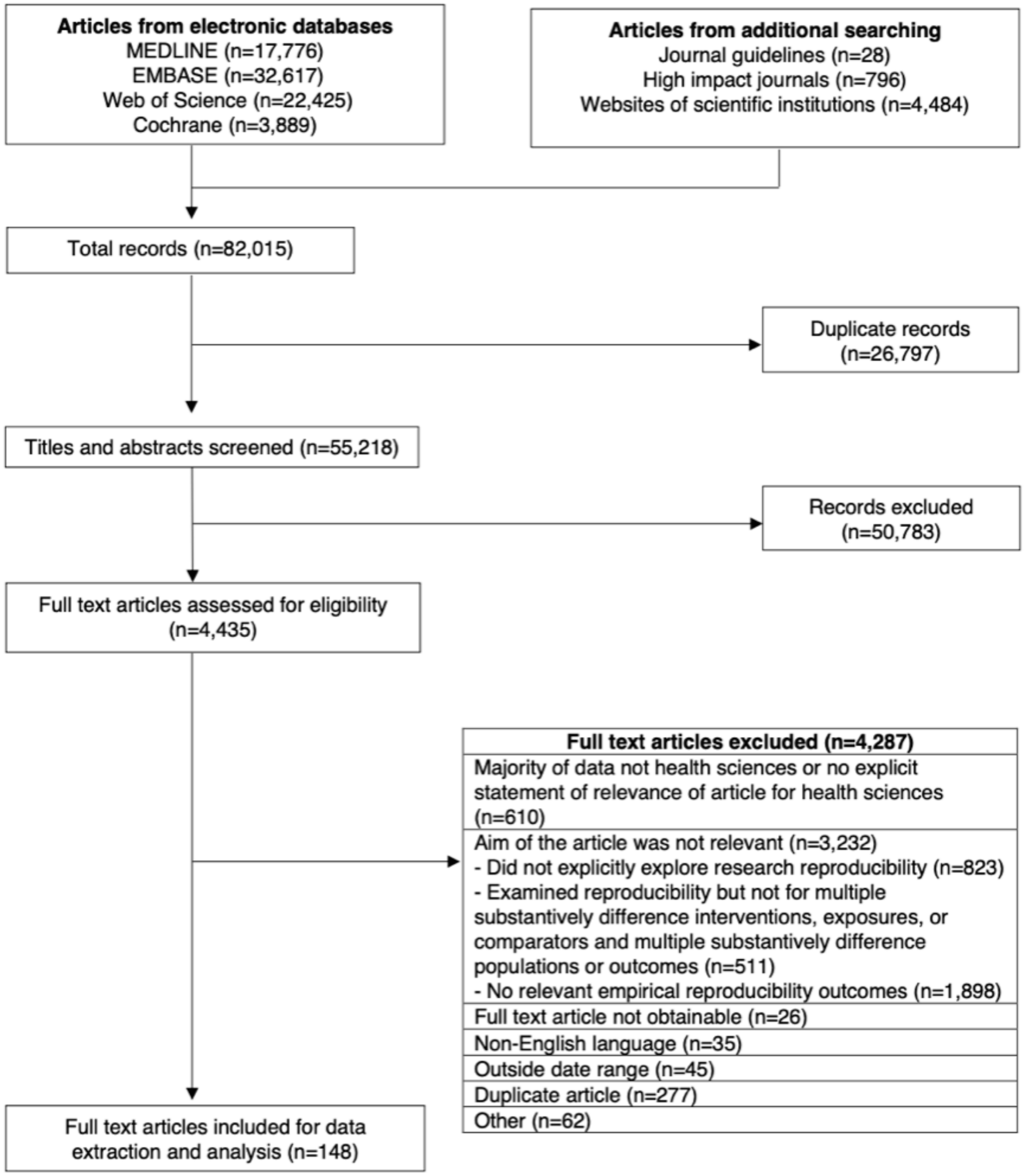

Figure 1. PRISMA flow diagram for study selection process.

We summarized risk of bias by domain to preserve transparency across heterogeneous designs.

2.5. Data synthesis

We used qualitative content analysis to label and group factors and then summarized the distribution of factor categories and reproducibility outcomes using evidence and gap maps.

First, we labeled factors using content analysis [24]. We used an inductive approach to synthesize data on factors in the form of short descriptive codes without imposing preconceived theoretical perspectives. The included studies were independently read in duplicate by two authors (JK, SJM) to understand the content and identify units of meaning. These units of meaning were then labeled with descriptive codes. Codes were aggregated by theme into factor categories. All codes were classified as modifiable or nonmodifiable based on the ability of investigators to act on the variable being described (eg, reporting of methods [modifiable] vs journal impact factor [nonmodifiable]) (Supplementary Table 5).

Table 1. Included studies by health sciences domain and reproducibility outcome

Health sciences domain	Total studies (N = 148, 100%)	Methods (N = 52, 36%)	Results (N = 96, 65%)
Biomedical/preclinical	62 (42%)	15 (30%)	46 (48%)
Clinical	71 (48%)	31 (58%)[a]	41 (43%)[a]
Health/social care services	1 (1%)	0 (0%)	1 (1%)
Population/public health	2 (1%)	0 (0%)	2 (2%)
Multiple	12 (8%)	6 (11%)	6 (6%)

Most studies were in biomedical/preclinical and clinical domains, and assessments of results reproducibility were more frequent than methodological reproducibility.
[a] Two studies assessed both methods and results reproducibility.

Second, we conducted evidence and gap maps to describe the distribution of factor categories and reproducibility outcomes by research discipline and study design [25—28]. We structured the maps as matrices in which rows represent factor categories and columns represent health research domains and study designs, and we present these visually as heat-map-style figures [29—31]. Data synthesis was performed using STATA-IC (V16), GraphPad Prism (V10), and Microsoft Excel (V16.83).

3. Results

We screened 55,218 title and abstracts and included 148 studies (Fig 1). Summary study characteristics are reported in Supplementary Table 6. The data are presented as follows: 1) description of included studies; 2) summary of factors evaluated and their distribution across health research domains and study designs; and 3) synthesis of modifiable and nonmodifiable factors and their associations with methodological or results reproducibility. Most studies examined biomedical/preclinical (62, 42%) or clinical (71, 48%) research. Methodological reproducibility was measured in 52 studies (35%), and results reproducibility was measured in 96 studies (65%) (Table 1).

3.1. Factors

There were 234 factors reported in the 148 studies with a mean of 1.5 factors reported per study (standard deviation: 1.1) (Table 2). Most factors were reported from observational studies (145 times, 65%) and evidence syntheses of observational studies (75 times, 34%). Three factors (1%) were reported from two experimental studies (Supplementary Table 7). Seventy-six factors (32%) were on methodological reproducibility and 158 factors (68%) were on results reproducibility (Table 2). In total, statistical tests of associations between factors and reproducibility outcomes were reported for 155 of 234 factors. The empirical data are summarized in Supplementary Table 8.

The factors were classified into 12 modifiable and three nonmodifiable categories. Modifiable factors were reported in 121 studies (80%), nonmodifiable factors were reported in seven studies (6%), and both modifiable and nonmodifiable factors were reported in 20 studies (14%) (Supplementary Table 6). Four modifiable factor categories: *transparency and reporting* (38, 16%), *analytical methods* (34, 14%), *study rigor and quality* (27, 11%), and *sample size and power* (25, 11%) represented over half of the factors reported. Evidence and gap maps showed that the number of studies reporting on each factor varied by health sciences research domain and study design (Fig 2).

Table 2. Distribution of factor assessments by domain and outcome

Health sciences domain	Total factors (N = 234, 100%)	Methods (N = 76, 32%) Modifiable (N = 63, 83%)	Methods (N = 76, 32%) Nonmodifiable (N = 13, 17%)	Results (N = 158, 68%) Modifiable (N = 126, 80%)	Results (N = 158, 68%) Nonmodifiable (N = 32, 20%)
Biomedical/preclinical	96 (41%)	17 (27%)	4 (31%)	64 (51%)	11 (34%)
Clinical	115 (49%)	40 (63%)	8 (62%)	50 (40%)	17 (53%)
Health/social care services	1 (1%)	0 (0%)	0 (0%)	0 (0%)	1 (3%)
Population/public health	8 (3%)	0 (0%)	0 (0%)	5 (4%)	3 (9%)
Multiple	14 (6%)	6 (10%)	1 (8%)	7 (6%)	0 (0%)

Empirical assessments most often evaluated modifiable factors for methods and results reproducibility in biomedical/preclinical and clinical domains; evidence remains relatively sparse in health/social care and population/public health domains.

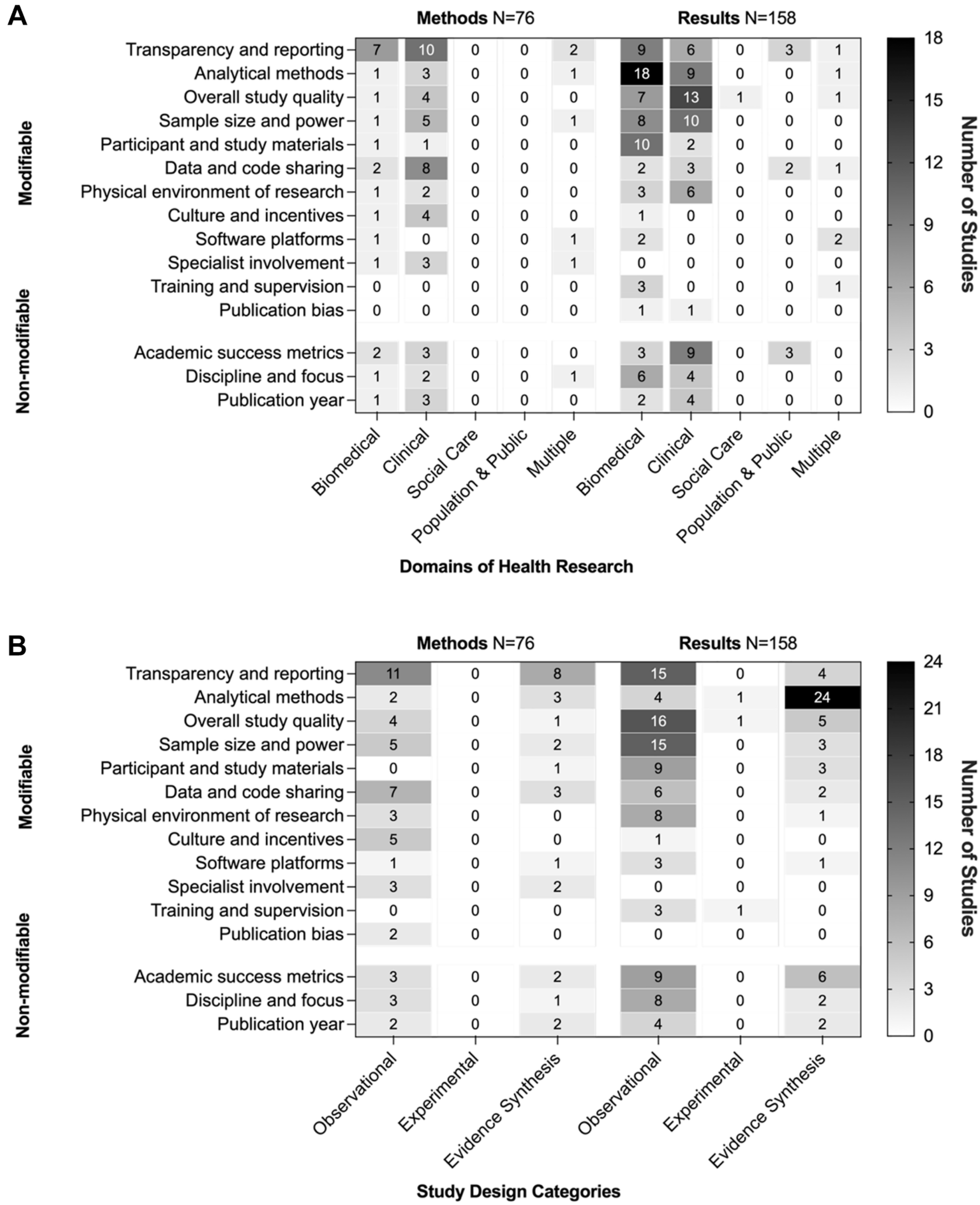

Figure 2. Evidence and gap maps of the number of studies reporting modifiable and nonmodifiable factors (rows) associated with the reproducibility of health sciences research outcomes (columns) by (A) health research domain and (B) study design. Numbers reported indicate number of included studies.

3.2. Modifiable factors

The evidence and gap maps identified five factor categories with 10 or more studies. In biomedical/preclinical research, *analytical methods* and *participants characteristics and study materials* were examined as factors associated with results reproducibility in 18 studies and 10 studies, respectively. In clinical research, the association between *transparency and reporting* and methodological reproducibility was examined in 10 studies, whereas *study rigor and quality* and *sample size and power* were examined as factors associated with results reproducibility in 13 studies and 10 studies, respectively.

Statistical tests of association between modifiable factors and reproducibility were reported 114 times. The factor categories most frequently evaluated were *analytical*

Table 3. Semiqualitative summary of empirical assessments reporting statistical associations between factors and methodological or results reproducibility

	Reproducibility outcomes		
	Total frequency	Methods	Results
Factor category	***N* = 234 (100%)**	***N* = 76 (32%)**	***N* = 158 (68%)**
Modifiable factors	*N* = 114 (49%)	*N* = 28 (37%)	*N* = 86 (54%)
Transparency and reporting	↕ **9**	↕ 5	↕ 4
N = 38 (16%)	↔ **5**	↔ 2	↔ 3
Analytical methods	↕ *24*	↕ 3	↕ *21*
N = 33 (14%)	↔ *2*	↔ 0	↔ *2*
Study rigor and quality	↕ **12**	↕ 1	↕ **11**
N = 27 (11%)	↔ **5**	↔ 1	↔ **4**
Sample size and power	↕ *21*	↕ 6	↕ *15*
N = 25 (11%)	↔ *2*	↔ 0	↔ *2*
Data and code sharing	↕ 1	↕ 0	↕ 1
N = 18 (7%)	↔ 2	↔ 0	↔ 2
Participants characteristics and study materials	↕ *10*	↕ 2	↕ *8*
N = 14 (6%)	↔ *2*	↔ 0	↔ *2*
Physical research environment	↕ 6	↕ 2	↕ 4
N = 12 (5%)	↔ 3	↔ 1	↔ 2
Culture and incentives	↕ 0	↕ 0	↕ 0
N = 6 (3%)	↔ 1	↔ 1	↔ 0
Software platforms	↕ 1	↕ 0	↕ 1
N = 6 (3%)	↔ 1	↔ 0	↔ 1
Specialist involvement	↕ 3	↕ 3	↕ 0
N = 5 (2%)	↔ 1	↔ 1	↔ 0
Training and supervision	↕ 0	↕ 0	↕ 0
N = 4 (2%)	↔ 1	↔ 0	↔ 1
Publication bias	↕ 2	↕ 0	↕ 2
N = 2 (1%)	↔ 0	↔ 0	↔ 0
Nonmodifiable factors	*N* = 41 (18%)	*N* = 11 (14%)	*N* = 30 (19%)
Academic success metrics	↕**11**	↕ 2	↕ **9**
N = 20 (9%)	↔ **8**	↔ 2	↔ **6**
Research discipline	↕ **8**	↕ 3	↕ **5**
N = 14 (6%)	↔ **6**	↔ 1	↔ **5**
Publication year	↕ 4	↕ 2	↕ 2
N = 10 (4%)	↔4	↔ 1	↔ 3

Sorted by frequency of total factors. Counts reflect analyses of study-defined units (participants or studies) within each study. Arrow symbols denote the direction of the statistical finding (↕ = at least one statistically significant association reported; ↔ = no statistically significant association reported); they do not imply causal effects or pooled effect sizes.

Individual studies could have assessed more than one factor category. Units' follow the unit of analysis defined by each included study (e.g., participants or studies). Findings should be interpreted with consideration of risk of bias (Supplementary Table 9) and the evidence and gaps maps (Fig 2), which contextualize where evidence is abundant or sparse. For example, if analyses rely on flexible analytic choices and the risk of bias domain for selective reporting is frequently "high/unclear," readers should treat nominal "significance" with caution.

N, number of studies; ↕ denotes number of studies that assessed for associations and found a statistically significant association; ↔ denotes number of studies that assessed for associations and did not find a statistically significant association.

Bold values indicate factors for which ≥10 studies assessed for associations; italics values indicate factors for which ≥10 studies assessed for associations and >75% of these studies showed a consistent effect.

methods (26 studies, 23%), *sample size and power* (23 studies, 21%), *study rigor and quality* (17 studies, 15%), and *participants characteristics and study materials* (12 studies, 11%) (Table 3). More studies reported statistical tests of association for results reproducibility (86 studies, 74%) than methods reproducibility (28 studies, 25%) (Table 3).

For results reproducibility, 21 of 23 studies (91%) that evaluated the statistical association between *analytical methods* and results reproducibility reported statistically significant findings. These studies were primarily evidence syntheses (17 evidence syntheses, 81%; Fig 3). Specific *analytical methods* factors that were reported by multiple studies to be associated with increased results reproducibility included statistical correction procedures (four studies, 15%), using more stringent statistical thresholds (eg, $P < .005$ compared to $P < .05$; three studies, 12%), and P value calibration techniques (three studies, 12%). In addition, systematic reviews that identified articles using a peer-reviewed search strategy, as well as systematic reviews that included a meta-analysis (compared to systematic reviews that did not include a meta-analysis), were associated with higher methodological reproducibility.

Fifteen of 17 studies (88%) that evaluated the statistical association between *sample size and power* and methodological or results reproducibility reported statistically significant findings. Sufficiently powered samples were associated with increased methods and results reproducibility.

Eleven of 15 studies (73%) that evaluated the association between *study rigor and quality* and results reproducibility reported statistically significant findings. Specifically, studies reported that quality of reporting (eg, following CONSORT guidelines, accuracy of methodology, and completeness of reporting), methodological independence from prior investigations, allocation concealment, and reporting forest plots with numerical point estimates (compared to forest plots without numerical point estimates) in meta-analyses were statistically associated with increased results reproducibility. One study found that seeking or having FDA approval was associated with improved methodological reproducibility among 220 prospective clinical trials. In contrast, inadequate and inaccurate reporting of study methodology was associated with decreased results reproducibility. Two studies found no association between trial randomization and reproducibility of methods or results.

Ten of 14 studies (71%) that tested the association between *participants characteristics and study materials* and reproducibility reported statistically significant findings. Specifically, increased inter-study variability including variability in setting, materials, experimental methods, and genetic variations in subjects was associated with decreased results reproducibility. One study found that increased researcher expertise was associated with increased forecasting accuracy of results reproducibility. However, increased researcher confidence was associated with lower forecasting accuracy. Finally, one study found that Cochrane reviews had higher methodological reproducibility of search strategies compared to non-Cochrane reviews. Further details of empirical evidence for each study are found in Supplementary Table 8.

3.3. Nonmodifiable factors

The evidence and gap maps identified that none of the nonmodifiable factor categories had 10 or more studies in a health sciences research domain or study design category. No more than three studies reported assessing any given nonmodifiable factor category for methods reproducibility. For results reproducibility, there were nine studies that measured *academic success metrics* in clinical research and six studies that measured *research discipline* in biomedical and preclinical research.

Statistical tests of association between nonmodifiable factors and reproducibility were reported 41 times. The factor categories most frequently evaluated were *academic success metrics* (20 studies, 9%) and *discipline focus* (14 studies, 6%) (Table 3). Among the 20 studies evaluating *academic success metrics*, eight studies (40%) examined the association between journal impact factor and reproducibility. The findings on this association were mixed: three studies (38%) found no association between journal impact factor and methods reproducibility (two studies) or results reproducibility (three studies), while five studies (62%) found that higher journal impact factor was associated with increased methods reproducibility (two studies) and results reproducibility (three studies). Overall, nine of 15 studies (60%) that tested the association between *academic success metrics* and results reproducibility reported statistically significant findings.

Five of 10 studies (50%) that tested the association between *research discipline* and results reproducibility reported statistically significant findings. The results from these studies were also mixed: three studies reported a statistically significant association between journal/subject area and methodological reproducibility. Three studies reported a statistically significant association between journal/subject area and results reproducibility, while three studies found no association with results reproducibility (Supplementary Table 8).

3.4. Risk of bias

Most studies were rated at low risk of bias for study participation, factor measurement, outcome measurement, and statistical analysis categories. Most studies were rated at moderate or high risk of bias for study confounding, attributable to missing or insufficient consideration to potential bias due to confounding in the design or analysis phase of the study (Supplementary Table 9). Association counts (Table 3) and EGMs (Fig 2) should be interpreted alongside risk of bias profiles.

4. Discussion

This systematic review and evidence and gap map synthesized data from 148 studies and reported associations between 234 factors and the reproducibility of health sciences research. Two-thirds of studies focused on the reproducibility of results, and nearly all available evidence came from biomedical/preclinical research or clinical research domains. Most studies used an observational study design. *Analytical methods*, *sample size and power*, and *participants characteristics and study materials* were categories

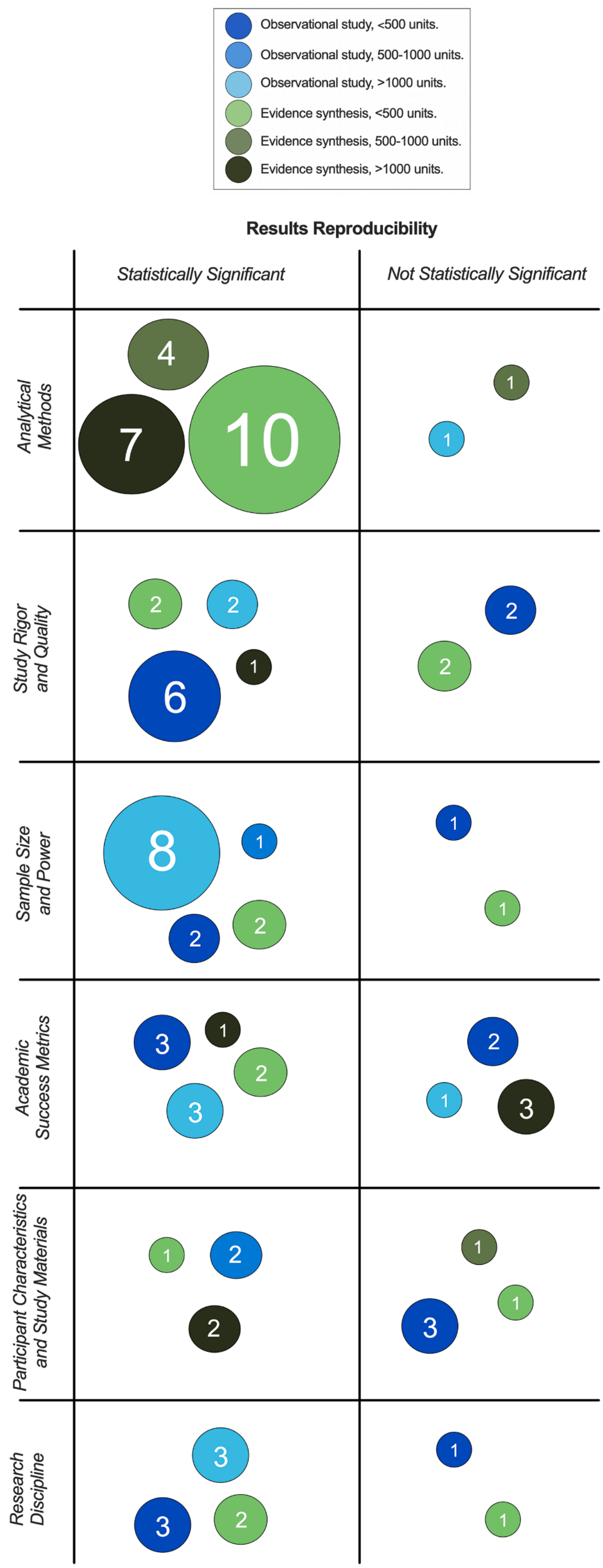

Figure 3. Evidence and gap map of factors categories for which 10 or more studies assessed for associations with results reproducibility. Results reproducibility reported by study design (observational study or evidence synthesis) and units of analysis (eg, participants or research studies) as defined by the study; circle size is proportional to the number of studies.

of factors most frequently and consistently reported to be associated with reproducibility.

A 2016 survey by *Nature* reported that most scientists perceived a ‘‘reproducibility crisis’’ in science [5]. Among the 1576 researchers surveyed, 60% of respondents stated that problems with reproducibility could be attributed to pressures to publish and selective reporting. Additional challenges included low statistical power or poor analysis, lack of replication within original laboratories, and insufficient mentoring. Other publications, primarily expert opinions, have provided complementary perspectives [3,9,10,32−34]. Our findings are broadly consistent with prior field-specific reviews, which have highlighted threats to reproducibility from low statistical power, selective or incomplete reporting, and suboptimal analytical practices [18−21].

Our study offers five key findings. First, using more stringent statistical testing procedures and thresholds was associated with greater results reproducibility. This included consideration of imputation methods [35−37], correcting for the Winner's curse [38], and applying false discovery rate-controlling procedures [39]. Implementing more stringent P value thresholds (eg, $P < .005$ compared to $P < .05$) [40] and using raw P values instead of calibrated P values were associated with fewer false positive findings [41−44]. Second, appropriate and sufficiently powered samples were associated with increased results reproducibility [35,36,39,45−55]. Among randomized controlled trials, consistent effects reported by subsequent trials were more likely when the initial trial had a larger sample size and higher statistical power [51,56]. Third, transparency and completeness of research reporting was associated with methodological reproducibility. Availability of study material and sampling methodology [57,58], higher PRISMA score (based on compliance to the PRISMA checklist) [59], and reporting sample size calculations [60] were associated with increased methodological reproducibility. Several studies identified that methodological and analytical procedures, adhering to reporting guidelines, and reporting sufficient elements of study methods and analyses for study replication were associated with increased results reproducibility [61−63]. Additional research is needed to clarify our findings that systematic reviews with meta-analyses and Cochrane reviews had superior methodological reproducibility than reviews without meta-analyses or non-Cochrane reviews, respectively. These findings may reflect differences in data sources, study designs, journal reporting parameters, or study-specific methodological rigor and transparency. We cannot infer from our data that meta-analyses or Cochrane reviews have an inherent advantage over their counterparts. Fourth, differences in study *participants characteristics and study materials* such as variations in specimens, genes, and laboratory settings were associated with decreased results reproducibility [64−67]. Fifth, there was limited evaluation and/or inconsistent associations between *data and code sharing*, *academic success* *metrics* (ie, citation number and journal impact factor), and *training/supervision* with reproducibility. Finally, few studies examined reproducibility within health/social care services and population/public health.

4.1. Limitations

The primary limitation of this review is that the heterogeneity in study methods, designs, domains of health sciences research, and types of reproducibility among the studies prevented a quantitative synthesis of the evidence. We identified factors for which there is consistent evidence of associations with reproducibility, but we were not able to estimate the magnitude of associations using meta-analysis.

Another limitation is that our choice of bibliographic databases may have influenced the studies we identified in our search. We purposefully focused on multidisciplinary databases with broad coverage of health sciences. While this approach facilitated feasibility and broad relevance to health sciences, we may have under-ascertained studies in select fields (eg, psychology). Furthermore, the English-language restriction may have excluded relevant non-English studies. We interpreted gaps in the literature cautiously, recognizing that they may reflect limitations of database coverage or potential language effects.

4.2. Recommendations and future directions

Based on the available data, there are three concrete recommendations for future directions to improve the reproducibility of health sciences research. First, experimental studies are needed to test interventions to improve reproducibility. The available data are almost entirely observational and provide plausible hypotheses that are worth testing with experimental designs. The use of more stringent statistical procedures and thresholds, sample size and power calculations, and quality assessments are reasonably low-risk, methodological approaches to implement pending further data. Nonetheless, there is likely a trade-off between the efficiency and reproducibility of science [68]. Embedding methodological experiments within health sciences studies would help us to understand any trade-offs and how to position science to evolve effectively and efficiently. Second, more empirical research on reproducibility is needed across the domains of health/social care services and population/public health. Given that evidence from studies within these domains directly informs health care and health system decisions, it is noteworthy that little is known about the factors that affect the reproducibility of this research. Addressing this gap is important for patient and population health and to improve public confidence in the health sciences. Third, we must better understand how structural/institutional factors influence research reproducibility. Such factors include the physical environment in which research is conducted, research culture and incentives, and research training and supervision. These elements shape scientific trends and day-to-day practices.

5. Conclusion

Our systematic review identified a large body of primarily observational data that suggest that reproducibility may be improved by implementing more stringent statistical testing procedures and thresholds, sample size and power calculations, and improved transparency and completeness of reporting. Experimental studies are now needed to test these hypotheses. Improvements to research reproducibility will only be achieved through large-scale shifts within the scientific community. Such changes will require coordination, time, and intentional action. Thus, we urge investigators to use rigorous and transparent methods, journals and reviewers to enforce reporting standards and ensure adequate space for description of methods, and funders and institutions to incentivize reproducibility through policy, training, and recognition.

CRediT authorship contribution statement

J. Kennett to Juliane Kennett: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **S.J. Moss to Stephana Julia Moss:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **J. Parsons Leigh to Jeanna Parsons Leigh:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **N. Bobrovitz to Niklas Bobrovitz:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **H.T. Stelfox to Henry T. Stelfox:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare no competing interests.

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Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2026.112135.

Data availability

All data used is publicly available from included studies.

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